# Uri Dickman

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## Education

Sept 2025 – Present
**University of California, Santa Barbara**
*Santa Barbara, CA*
Doctor of Philosophy, Mechanical Engineering
Certificate in College and University Teaching
Fields: Computational Science and Engineering; Solid Mechanics, Structures, and Materials

Jan 2025 – May 2025
**Case Western Reserve University**
*Cleveland, OH*
Visiting Graduate Student, Applied Mathematics
Coursework: Computational Neuroscience

May 2024
**Brown University**
*Providence, RI*
Bachelor of Science, Physics with Honors

## Research

Sept 2025 – Present
**Graduate Student Researcher**
*Computational Applied Science Laboratory*
Department of Mechanical Engineering, UC Santa Barbara
Santa Barbara, CA
Project: Mesoscale Operator Surrogate for Alloy Interface Crystallization
Advisor: Frederic Gibou

July 2024 – Aug 2025
**Project Analyst**
*Computational Biomathematics Laboratory*
Department of Applied Mathematics, Case Western Reserve University
Cleveland, OH
Project: Fast, flexible, Python-integrated simulation of biophysical neural networks with complex plastic synapses
Supervisor: Peter Thomas

Jan 2023 – May 2024 **Undergraduate Research Assistant**
*Solid State Nanofluidics and Nanoionics Group*
Department of Physics, Brown University
Providence, RI
Project: The Specific Heat of Nano-confined Fluids
Advisor: Matthias Kuehne

## Teaching

Sep 2025 – Present **Graduate Teaching Assistant**
Dept. of Mechanical Engineering, UC Santa Barbara
*Santa Barbara, CA*

- Engineering Mechanics: Dynamics (ME 16)
- Electrical and Electronic Circuits (ME 6)
- Mathematics of Engineering (ME 17)

Sep 2021 – Aug 2024 **Undergraduate Teaching Assistant**
Dept. of Physics & Dept. of Mathematics, Brown University
*Providence, RI*

- Calculus I (MATH 90)
- Calculus III (MATH 200)
- Basic Physics A (PHYS 30)
- Basic Physics B (PHYS 40)
- Analytical Mechanics (PHYS 70)

## Awards and Fellowships

June – Aug 2026 **Space Vandenberg Graduate Student Mentor**
UCSB Center for Science & Engineering Partnerships
*Vandenberg Space Force Base, Santa Barbara, CA*

Feb – April 2026 **UR2PhD Mentorship Fellow**
Computing Research Association
*Remote*

May – Aug 2023 **Undergraduate Teaching and Research Award**
SPRINT, Brown University
*Providence, RI*

Sept – Dec 2022 **Mathematics Teaching Fellow**
Department of Mathematics, Brown University
*Providence, RI*

## Presentations

November 19, 2025 **Society for Neuroscience Annual Meeting**
*San Diego, CA*

- Presenting author, NEURONpyxl: Fast, flexible, Python-integrated simulation of biophysical neural networks with complex plastic synapses
- First-author, Fast/slow dissection and dimension reduction for a model of pattern generator variability
- Co-author, Modeling mechanisms of pattern generator variability

April 30, 2024 **Senior Thesis Oral Defense**
*Providence, RI*
Title: The Specific Heat of Nano-confined Fluids

August 6, 2023 **Summer Research Symposium**
*Providence, RI*
Presenting author, The Specific Heat of Nano-confined Fluids

## Professional Memberships

Jan 2025 – Jan 2026 **Society for Neuroscience**
Graduate Student Member

May 2024 – May 2025 **Sigma Xi Honor Society**
Associate Member

## Technical Skills

*Programming languages*
Python, C++, C, MATLAB, Mathematica, Julia

*Scientific tools*
PyTorch, Slurm, Linux, MPI, Git, NEURON, LAMMPS, LaTeX, Zotero

*Skills*
Scientific & high-performance computing, numerical methods, deep learning, research methods, mentorship, engineering education

## Selected Projects

**Self-hosted High-performance Computing Cluster**

- Includes 9 Raspberry Pi nodes and 1 NVIDIA GPU node
- Slurm used for resource allocation
- Tunneling technology used to enable remote SSH login
- Advanced system administration for account and file management

## Editorial Contributions

January 2026 – Present **Peer Reviewer**
Journal of Computational Physics

## Publications

Dickman, U., Thomas, P. J., Chiel, H. J., Byrne, J. H., and Neveu, C. L. (2026). Neuronpyxl: fast, flexible, python-integrated simulation of biophysical neural networks with complex plastic synapses. *Frontiers in Computational Neuroscience*, Volume 20 - 2026

Peng, C., Ginzburg, J., Dickman, U., Bair, J., and Kuehne, M. (2025). Thermal characterization of suspended fine wires across continuum to free-molecular gas regimes using the $3\omega$ method. *Phys. Rev. Appl.*, 24:064060